

ANANYA SHAH

AI/ML Engineer | Software Developer

Highly motivated **Software Developer** and **AI/ML Engineer** with strong experience in **Machine Learning, Generative AI, AWS Cloud**, and full-stack development. Recognized for outstanding innovation by being **Selected for the Dewang Mehta IT Awards 2025** (Top 14 Projects) for the Gen-AI Powered Quotation System. Seeking to leverage deep technical expertise in advanced studies focusing on Artificial Intelligence and Large Language Models.

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EDUCATION

Ahmedabad Institute of Technology

Bachelor of Engineering (BE) in Computer Engineering

2021 – 2025

IIT Roorkee & Masai School

Specialization Program in Artificial Intelligence & Machine Learning

April 2026 - Present

SKILLS

- **Languages:** Python, JavaScript, SQL, C, HTML, Java, .NET, TypeScript, CSS
- **AI/ML:** NLP, LLMs, Deep Learning, Vector Databases, SentenceTransformer, PyTorch, Scikit-Learn, RAG, Reinforcement Learning (DQN, Q-Learning), OpenCV, Prompt Engineering
- **Frameworks:** Node.js (Express.js), Next.js, React.js, Flask, NumPy, Pandas, FastAPI, Tailwind CSS
- **Tools:** Git, GitHub, REST APIs, Docker, Jupyter Notebook, VS Code, Google Colab, Postman
- **Database:** SQL Server, MongoDB, Oracle, Firestore, Redis
- **Cloud Technology:** AWS (S3, EC2, CodePipeline, CodeBuild, ECR, CLI, Secret Manager, IAM Service)
- **Soft Skills:** Problem Solving, Team Collaboration, Client Communication, Requirement Gathering & Analysis, Project management, Issue Escalation & Resolution, Cross-Functional Team Leadership, Analytical Problem Solving, Agile Collaboration

WORK EXPERIENCE

OPERISOFT TECHNOLOGIES PVT. LTD.

Software Developer | Full-time

June 2025 – Present

- **System Architecture & Full-Stack:** Develop and maintain both Microservice and Monolithic web architecture using Node.js, Python, Next.js, and React.js, building high-performance RESTful APIs and responsive user interfaces.
- **Cloud & Containerization:** Containerize services using Docker and implement Redis caching to optimize data retrieval latency; deploy and manage cloud infrastructure across AWS (S3, EC2, CodePipeline, CodeBuild, ECR, CLI, Secret Manager, IAM Service).
- **Agile Workflow & Client management:** Practice Agile/Scrum methodologies, actively leading client communication, requirement gathering, and technical issue escalations to align sprint deliverables with business needs.
- **Mobile & Android Performance:** Support Android app development by building modular UI components, components, integrating backend REST APIs, and resolving performance bottlenecks to enhance app stability.

CYGNET.ONE

AI/ML Intern

Jan 2025 – Apr 2025

- **Award-Winning Innovation:** Engineered the Gen-AI Powered Quotation Generation System, recognized as a Top 14 Finalist at the Dewang Mehta IT Awards 2025.
- **Gen-AI & Vector Search:** integrated embedding-based semantic search (Sentence Transformer) with Vector Databases in a Flask and React production pipeline, reducing manual quotation creation time by 80%.
- **Workflow Automation:** Built dynamic PDF rendering engines and tier classification algorithms for instant quote export and automated pricing recommendations.

CYGNET.ONE

AI/ML Intern

May 2024 – Aug 2024

- **Information Retrieval System:** Architected an Insurance Document Search System using Python and Machine Learning algorithms for efficient text extraction and document indexing.
- **Feature Engineering:** Design automated feature extraction pipelines and semantic matching models, improving query search precision and retrieval accuracy.

CENTER OF EXCELLENCE (SAP COE)

Participant

2023 – 2025

- **Computer Vision & ML Projects:** Developed end-to-end Machine Learning models including Face Recognition and Object Detection using OpenCV and PyTorch.

- **Specialized Training:** Completed hands-on training in Deep language, Computer Vision, Cybersecurity, IOT, and SAP ABAP on SAP Business Technology Platform (BTP).

KEY PROJECT

GEN-AI POWERED QUOTATION GENERATION SYSTEM (Cygnet.One)

Dewang Mehta IT Awards 2025 – Selected (Top 14)

- **Generative AI & Vector Search:** Engineered an automated quotation platform leveraging OpenLLaMa 3B and Vector Database (Sentence Transformer) for semantic product search and dynamic pricing classification.
- **Full-Stack & Export Pipeline:** Built a robust Flask backend FastAPI connected to an interactive React.js + TailwindCSS dashboard featuring automated PDF generation via jsPDF.
- **Tech Stack:** Python (Flask), React.js, OpenLLaMa 3B, SentenceTransformer, Vector Database, jsPDF, TailwindCSS

PROJECT SUBMISSION & REVIEW PLATFORM

Full Stack & AWS Cloud Architecture

- **Automated CI/CD:** Configured end-to-end continuous deployment using AWS CodePipeline and CodeBuild, building container images, pushing to Amazon ECR, and executing remote deployments on eC2 using AWS Systems manager (SSM).
- **Cloud Security & Secrets:** Enforced security best practices using AWS IAM instance profiles (removing hardcoded keys in production) and AWS Secrets Manager for environment variables.
- **Storage & Routing:** Provisioned AWS S3 buckets for encrypted project artifact storage with presigned URL expiration policies, and configured Nginx as a reverse proxy for SSL termination and request forwarding.
- **Clean Full-Stack Architecture:** Structured the Next.js 14 frontend using an enterprise layered pattern - decoupling presentation, state, and API calls with Axios interceptor - and built a modular FastAPI backend separating routes, business logic, validation schemas, and database models.
- **Containerization & Service Isolation:** Authored multi-stage Dockerfiles and Docker Compose orchestrations leveraging Supervisor process control to run decoupled frontend and backend services concurrently behind an Nginx reverse proxy.

AUTOMATED RACING DEEP RL AGENT

Deep Reinforcement Learning & Neural Networks

- **Reinforcement Learning:** Trained a Deep Reinforcement Learning agent utilizing Deep Q-Networks (DQN) and PyTorch for autonomous track navigation, obstacle avoidance, and steering control in simulated environments.

- **Model Optimization:** implemented Q-Learning algorithms and continuous reward functions to optimize agent velocity, path selection, and decision-making accuracy.
- **Tech Stack:** Python, PyTorch, Deep Q-Networks (DQN), Reinforcement Learning. OpenAI Gym, Numpy

CERTIFICATIONS [↗](#)

- **Agentic AI, Generative AI & Reinforcement Learning** (Udemy) — Feb 2026 [↗](#)
- **British Airways Data Science Certification** (British Airways/Forage) — Feb 2026 [↗](#)
- **Deloitte Data Analytics Certification** (Deloitte/Forage) — Feb 2026 [↗](#)
- **Machine Learning, Deep Learning, Computer Vision & SAP ABAP on BTP** (Edunet Foundation - COE) — Apr 2024 [↗](#)
- **Data Analysis with Pandas and Python** (Udemy) — Nov 2025 [↗](#)
- **The Complete Python Programming Course** (Udemy) — Mar 2023 [↗](#)

Extra Curricular Activities

2021 – 2025

- Served as a member of the **Google Developer Student Club (GDSC)** for one year, contributing to technical and community-focused initiatives.
- Core team member for **Techfest 2024**, assisting in planning, coordination, and successful execution of technical events.
- Organized and participated in multiple college events, ensuring smooth event management and active involvement in campus activities.